Module 4 - Summary Video – Transcript
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Hello everyone and welcome. I am Maria Konte, the instructor of the Computer Networks course. In this video, we review AS relationships and interdomain routing. The main question that we focus on is how does traffic reliably move between independently run networks across the global Internet?

So inside a single network, routing, we can say, is mostly a technical problem because we are optimizing paths within one administrative domain. But between networks, routing also is a mix of technology problems and policy decisions because different organizations have different goals, constraints, and business relationships. So this is where the BGP protocol comes in. BGP, we can say, is the glue of the Internet because it connects networks together and gives them a way to express routing decisions that reflect real-world incentives, for example, which routes a network will accept and which routes it will share with the neighbors. And this matters because the Internet is not just one network, it is an entire ecosystem of thousands of independent networks that are cooperating but also competing, as we will see in a second.

In this video, we take a look at how traffic moves between networks.

So let's get started by looking at the scenario that we have on the picture and asking the question where does routing live in the Internet architecture stack? So here we have a host that runs a browser application inside an enterprise network.

The traffic that leaves from the host that runs the browser application leaves the enterprise network through routers, enters an Internet service provider network, and eventually reaches the content distribution network where the web server is hosted.

So that traffic is crossing multiple independently run networks. Also, taking a look at the protocol stack, we see that we have the HTTP protocol running on top of the TCP protocol and that over IP. So routing primarily sits at the IP layer. The IP layer is responsible for providing the protocols that move packets hop by hop across routers, and that is independent of which application is running at the top. So here let us also make one important note that inside the single network that is operated by one administrative domain, routing is handled by intradomain routing protocols that optimize internal paths. But across independently operating networks, routing needs a different approach because different networks, as we mentioned, have different policies and because interdomain routing also needs to scale across thousands of networks. Also, let us clarify what we refer to as a network at this level, or as we say, an autonomous system. So an autonomous system is a network that is owned and operated by a single organization. This can be an Internet service provider, a university, or an enterprise. So it is managed under one administrative authority, meaning that it controls its own infrastructure and allocated IP address space and follows a specific set of routing decisions.

Now, inside the AS, routers typically use the same internal routing protocol. It can be, for example, OSPF or RIP to select paths. While between ASes, the routers use BGP, the BGP protocol, to exchange routing information and decide how traffic should flow across organizational boundaries.

Let us also make one more note over here that one organization doesn't always mean one AS. For example, a large ISP might run multiple ASes for business separation, traffic engineering purposes, and so on.

So inside the AS, we use intradomain routing protocols to find good internal paths, but at the edges the AS makes interdomain choices or decisions, meaning which neighbors to connect to, which routes to announce, which routes to accept, which exits to prefer for traffic engineering purposes, and so on. Now that we have seen why routing splits into what is happening inside the network versus between networks, the next question that we ask is what exactly is exchanged between networks to make interdomain routing work. So at Internet scale, destinations are not individual hosts; they are actually entire address blocks, such as, for example, the prefixes that we see on this slide. So what networks exchange is actually reachability to those prefixes, to those address blocks. This reachability information is learned and advertised at the border routers where networks connect. For example, on this slide, the enterprise network learns about external reachability from the ISP edge, and the ISP learns about reachability to the content distribution address block, also through its border routers. Now inside its network, that external information is also shared so that internal routers know which exit to use.

So in summary, we can say that the end-to-end flow is as follows. The border routers first learn about reachability information about paths, essentially for external destinations. They spread that internally and forward packets hop by hop towards the chosen exit.

Now that we have seen what is the type of information that is exchanged between networks, the next question that we are asking is what are those networks that are actually performing this exchange of information and where do they meet to interconnect?

So we have different types of networks. We have Internet service providers across tiers, for example Tier 1, 2, and 3. We have enterprise networks. We also have content distribution networks, which offer better performance and cost, deploying distributed infrastructure closer to the users.

And also we have Internet exchange points. These are physical switching fabrics where networks meet to exchange traffic at scale. In practice, interconnection happens at points of presence and at IXPs.

Also, let us mention that historically the Internet had more of a hierarchical structure, with traffic often flowing up and down through large transit providers. But over time this has been flattening because more networks peer directly or interconnect at IXP facilities.

Now, this entire ecosystem is shaped by both competition and cooperation, because networks compete for customers, performance, and market share, and they do care about cost, latency, and control over traffic. But also they need to cooperate because no single network provides global connectivity on its own. So this tension between competition and cooperation creates the incentives behind routing policies. So more specifically, an AS will share routes and carry traffic when there is financial benefit.

And vice versa, it might restrict routes in other cases. So these incentives show up explicitly in AS business relationships. So let's review what are the types of those relationships. First, we have the customer-provider relationships, or also so-called transit.

In this relationship, the customer's role is to buy Internet reachability and connect their network to the provider. The provider's role, on the other hand, is to sell reachability and carry traffic between the customer and the rest of the Internet.

Let us also mention a key point over here that the customer pays the provider regardless of the traffic direction across the link that they share. Whether there is traffic that is crossing outbound from the customer to the Internet or inbound from the Internet back to the customer, the customer still uses the provider's infrastructure because this is what they're paying for. Therefore, the customer advertises its own address blocks to the provider and therefore to the rest of the Internet, and so the provider knows how the customer is reachable. In return, the provider advertises a large set of routes back to the customer, often essentially the provider's full routing table, so the customer knows how to reach external destinations, and the provider also advertises the customer's routes outward to other networks. Now the second business relationship is peering. With peering, two networks typically don't pay each other to exchange traffic because it benefits both sides. But the exchange is more limited than the transit relationship that we just described.

So each peer usually advertises routes to provide reachability for itself and for its own customers. A peer typically doesn't advertise routes that they learn from other peers or from providers, because that would turn the peering relationship into free transit.

So now that we have described incentives, let us see how those are encoded into rules, policies essentially. So this is where BGP import and export policies come into play. The simplest way that we can think about this is from an AS's perspective. What routes do I need to advertise to other ASes, and which routes out of all the routes that I receive do I prefer? Let's start with exporting, with export policies. When we advertise a route to a neighbor, we're basically saying, we're signaling to the neighbor to send traffic for this destination through our network. Therefore, we offer reachability for a specific destination. So the question is, do we want to export all routes that we know?

If we learned a route from a customer, then we export broadly, meaning to providers and to peers, because carrying traffic to our customers will generate revenue for our network. But if, on the other hand, we learn a route from a peer or a provider, we typically export it only to our customers. We don't export it to other peers or providers because that would mean that we would offer free service between nonpaying networks. Now let's talk about selecting which routes to import. So for the same destination, a router may learn multiple possible routes from their customers, from their peers, and from their providers. A router will not import them and keep forwarding them all. In contrast, they will rank them and select the best route, and that ranking usually happens according to cost.

First, the router prefers routes that they learn through their customers, then they prefer routes that they learn through their peers, and finally through their providers.

So now let's zoom into this path selection process. More specifically, once a router has multiple routes to the same destination, what is the process that the router follows to select one path? First, during import, the router applies specific import policies.

It can reject routes, it can tag them, or it can assign specific attributes that encode specific preferences. We're going to talk about example attributes in a second. Then, during the decision process, the router compares candidate routes attribute by attribute in a fixed order until they select a specific path and finally they install that selected path. So this becomes the route that it will actually use for forwarding to that specific destination, and at the end it applies the export policy. In other words, it decides what it will advertise to other neighbors.

So given the above, we observe that for interdomain routing, it is rarely about selecting the shortest path. A router may have several valid routes, and the path it ultimately selects is often the one that best matches policies that reflect financial and traffic engineering goals.

Now let's zoom into BGP attributes that are used to express policies, and we will talk about two important ones, namely LocalPref, the LocalPref attribute, and the MED attribute. So starting with LocalPref.

This is set inside our network, inside our AS, and we can consider that as a lever that we can use for deciding which exit we want to use for outbound traffic, for traffic that exits our network.

So in practice, operators assign LocalPref in ranges based on business relationships with neighboring ASes. Therefore, a network operator can assign higher LocalPref values to specific routes so that these are selected by the router during the path selection process.

And therefore these paths will be used when traffic exits the network. For example, routes that are learned by customers can be assigned higher LocalPref values, then routes that are learned by peers can be assigned lower values, and then finally routes that are learned by providers can be assigned even lower values.

Now another attribute is the MED attribute, and this is typically set by the neighbor to signal where they prefer traffic to enter their network, for example in cases where there are multiple interconnection points between two neighbors. It is essentially here a suggestion or a signal to the neighbor that if you are sending traffic my way, send it through this entry point to my network. But because, as we mentioned, this is more of a signal or a suggestion, it's not possible to be enforced, so the network that receives those routes with specific MED values can honor the suggestion or signal from their neighbor, or they can ignore it.

Now let's zoom into BGP mechanics and also two flavors that the BGP protocol has. So BGP neighbors establish a session over TCP by first exchanging so-called open messages to set up the connection.

And from there they share routing information. After this initial exchange, BGP operates incrementally. It is using update messages to announce new routes or withdraw old ones, and also periodically it sends keepalive messages to maintain the session.

Now, the BGP protocol has two flavors, eBGP or external BGP, and iBGP or internal BGP. Let us look at the difference between the two. So the eBGP protocol runs between routers in different ASes.

So this is where border routers learn routes from neighbors and advertise routes outwards, constrained by the import and export policies that we just described earlier. On the other hand, the iBGP or internal BGP protocol runs within the same AS.

And the job that it has is to take what the border routers learn from eBGP and disseminate it through the routers internally so that routers throughout the same AS know which exit to use for each external destination.

It typically runs as a full mesh of sessions between routers inside the AS to ensure routes propagate correctly.

Now let us note over here that iBGP doesn’t compute internal paths because computing internal paths is still the job of intradomain routing, as we have been mentioning. So it is the job of protocols such as OSPF, RIP, and so on.

Now let us discuss how interdomain and intradomain routing work together. More specifically, how a router combines the information that it learns from BGP and from internal gateway protocols, intradomain routing protocols, to put together the so-called forwarding information base.

We can think of the forwarding information base as the router’s first lookup table. So when a packet arrives, the router consults this table to learn where to send the packet next and what is the correct outgoing port. So we can think of this process as the following four steps. The router selects the best path. So for a destination prefix, it compares the BGP routes that it learns and chooses the best path through the path selection process that we previously saw.

Then the router identifies the egress point. So this best path determines which border router will be used to exit the AS, and this is set by the network operator. The router installs a forwarding entry in the table to ensure that the packet for this specific prefix should be sent towards the chosen exit. And then finally, at the fourth step, the router relies on the internal intradomain routing protocols to reach that exit, so it will use intradomain routing information to forward traffic across the AS to the border router and then to the next network.

Now let's turn our attention to operational problems. What can go wrong? So we can have errors, we can have misconfiguration. So these are mistakes or faults that lead to incorrect or unstable routing. They usually show up by bad route announcements, for example, advertising wrong prefixes or a very large number of prefixes, or as route instability or churn, where routes may be changing too quickly or they may get repeatedly announced or withdrawn. Now because of how BGP works, all of these misconfigurations, errors, and perhaps unintentional mistakes do not stay local. They spread as bursts of update messages and withdraw messages, and they can spread across the neighbors. So once that happens, the network can experience slow convergence and instability or even outages. This is why operators add guardrails. For example, they can control churn using mechanisms such as flap damping and update thresholds. These are mechanisms that temporarily suppress routes that keep flapping up and down, and this improves stability by preventing constant oscillations.

So now that we have covered what can go wrong in BGP and the operational controls that the network used to keep routing stable, the next question that we ask is where does interconnection actually happen at scale? In practice, a huge number of peering happens at shared physical meeting points, so-called Internet Exchange Points, or IXPs in short. An IXP typically provides a shared peering fabric, which is essentially an Ethernet switching network that lets many organizations, such as, for example, Internet service providers, content distribution networks, cloud providers, universities, and others, connect in one place and exchange traffic directly locally. To peer at an IXP, a network generally needs the following: first, a physical connection to the exchange fabric. This can be, for example, a port on the IXP switch. Also, it needs a routable interconnection setup, essentially a border router running BGP along with an autonomous system number and prefixes to advertise.

And finally, it needs membership and onboarding with the IXP, and this means essentially paying the fees, meeting technical requirements, getting the right VLAN setup, and so on. So once connected, networks can peer in two main ways.

Either through bilateral peering, where the network sets up separate BGP sessions with its specific peer that they would like to exchange traffic with, or through multilateral peering, and that happens through the IXP's route servers.

OK, so let's focus on how route servers work. So architecturally, the route server has one shared routing table, which we refer to as the master table, and then builds separate routing tables for each member based on export policies and import policies. So more specifically, when let's say member X announces a prefix P1, the route server first runs import filters to check that member X is allowed to advertise prefix P1. This helps to prevent route leaks and enforces exchange or other membership rules that may exist. If the route passes these checks, then the route server adds that to the master table. Then it runs export filters based on what each receiving member wants. For member Z, for example, these filters decide whether Z should receive the advertisement for that prefix P1. The result is Z's own routing view in its per-member table, and the route server advertises the selected route to Z.

So the main idea is simple: route servers scale peering by collecting routes and distributing them broadly while still preserving the policies that every participant has through filtering and through per-member different routing views.

Now finally, let us talk about the benefits that IXPs offer and the different services. So IXPs create value in two ways. They make it cheaper and faster to exchange traffic, and they make interconnection easier to manage at scale.

There are the standard benefits such as, for example, lower transit costs, lower latency, higher bandwidth, but there are also additional reasons for networks to connect. An important reason is that it provides low-friction peering because many IXPs offer route servers as a free service.

So a network can start exchanging routes with many peers quickly without having to negotiate or maintain lots of separate bilateral agreements. In addition, instead of building and operating a BGP session with everyone, a member can peer once with a route server and still keep a few direct bilateral sessions for special or high-volume partners. Also, route servers can support resilience because they may provide the fallback path if a direct bilateral session goes down. IXPs also offer additional services such as, for example, remote peering via transport partners, therefore helping smaller or geographically distant networks to participate without having a full local presence. Also, many Internet exchange points provide security and community services, such as, for example, customer-triggered blackholing for DDoS mitigation.

Also let us mention that IXPs act as neutral marketplaces because public peering is typically settlement free and it is not priced per traffic volume, while networks can still choose different types of relationships, for example, public peering, paid peering, or transit over the same connection fabric.

So in summary, in this video, we learned how routing works between independently run networks on the Internet. We talked about autonomous systems. We talked about this contrast between intradomain routing or internal gateway protocols that are running inside an AS and the interdomain routing protocol such as BGP that runs between ASes. We also considered business relationships such as, for example, transit and peering. We also talked about BGP import and export policies.

We also saw how routers choose routes using different attributes such as, for example, LocalPref and MED. And finally, we talked about misconfiguration errors and instability that can happen, and we wrapped up with how IXPs and route servers work to offer not only interconnection services but also additional services.

Thank you for watching!